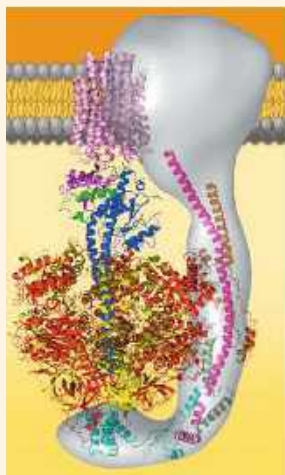


3. The final electron acceptor of the electron transport chain that functions in aerobic oxidative phosphorylation is
 - (A) O_2 .
 - (B) water.
 - (C) NAD^+ .
 - (D) pyruvate.
4. In mitochondria, exergonic redox reactions
 - (A) are the source of energy driving prokaryotic ATP synthesis.
 - (B) provide the energy that establishes the proton gradient.
 - (C) reduce carbon atoms to carbon dioxide.
 - (D) are coupled via phosphorylated intermediates to endergonic processes.

Levels 3-4: Applying/Analyzing

5. What is the oxidizing agent in the following reaction?

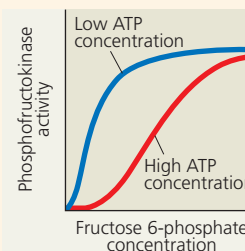
$$\text{Pyruvate} + \text{NADH} + \text{H}^+ \rightarrow \text{Lactate} + \text{NAD}^+$$
 - (A) oxygen
 - (B) NADH
 - (C) lactate
 - (D) pyruvate
6. When electrons flow along the electron transport chains of mitochondria, which of the following changes occurs?
 - (A) The pH of the matrix increases.
 - (B) ATP synthase pumps protons by active transport.
 - (C) The electrons gain free energy.
 - (D) NAD^+ is oxidized.
7. Most CO_2 from catabolism is released during
 - (A) glycolysis.
 - (B) the citric acid cycle.
 - (C) lactate fermentation.
 - (D) electron transport.
8. **MAKE CONNECTIONS** Step 3 in Figure 9.8 is a major point of regulation of glycolysis. The enzyme phosphofructokinase is allosterically regulated by ATP and related molecules (see Concept 8.5). Considering the overall result of glycolysis, would you expect ATP to inhibit or stimulate activity of this enzyme? Explain. (Hint: Make sure you consider the role of ATP as an allosteric regulator, not as a substrate of the enzyme.)
9. **MAKE CONNECTIONS** The proton pump shown in Figures 7.18 and 7.19 is a type of ATP synthase like that in Figure 9.13. Compare the processes shown in the three figures, and say whether they are involved in active or passive transport (see also Concepts 7.3 and 7.4).
10. **VISUAL SKILLS** This computer model shows the four parts of ATP synthase, each part consisting of a number of polypeptide subunits (the solid gray part is still an area of active research). Using Figure 9.13 as a guide, label the rotor, stator, internal rod, and catalytic knob of this molecular motor.



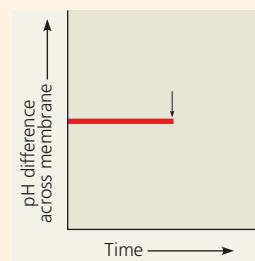
Levels 5-6: Evaluating/Creating

11. INTERPRET THE DATA

Phosphofructokinase is an enzyme that acts on fructose 6-phosphate at an early step in glucose breakdown. Regulation of this enzyme controls whether the sugar will continue on in the glycolytic pathway. Considering this graph, under which condition is phosphofructokinase more active? Given what you know about glycolysis and regulation of metabolism by this enzyme, explain the mechanism by which phosphofructokinase activity differs depending on ATP concentration. Explain why it makes sense that regulation of this enzyme has evolved so that it works this way.



12. **DRAW IT** The graph here shows the pH difference across the inner mitochondrial membrane over time in an actively respiring cell. At the time indicated by the vertical arrow, a metabolic poison is added that specifically and completely inhibits all function of mitochondrial ATP synthase. Draw what you would expect to see for the rest of the graphed line over the next short period of time. Explain.



13. **EVOLUTION CONNECTION** ATP synthases are found in the prokaryotic plasma membrane and in mitochondria and chloroplasts. (a) Propose a hypothesis to account for an evolutionary relationship of these eukaryotic organelles and prokaryotes. (b) Explain how the amino acid sequences of the ATP synthases from the different sources might either support or fail to support your hypothesis.
14. **SCIENTIFIC INQUIRY** In the 1930s, some physicians prescribed low doses of a compound called dinitrophenol (DNP) to help patients lose weight. This unsafe method was abandoned after some patients died. DNP uncouples the chemiosmotic machinery by making the lipid bilayer of the inner mitochondrial membrane leaky to H^+ . Explain how this could cause weight loss and death.
15. **WRITE ABOUT A THEME: ORGANIZATION** In a short essay (100–150 words), explain how oxidative phosphorylation—production of ATP using energy from the redox reactions of a spatially organized electron transport chain followed by chemiosmosis—is an example of how new properties emerge at each level of the biological hierarchy.

16. SYNTHESIZE YOUR KNOWLEDGE



Coenzyme Q (CoQ) is sold as a nutritional supplement. One company uses this marketing slogan for CoQ: “Give your heart the fuel it craves most.” Considering the role of coenzyme Q, critique this claim. How do you think this product might function to benefit the heart? Is CoQ used as a “fuel” during cellular respiration?

For selected answers, see Appendix A.